12. The Normal distribution is a limiting form of Binomial distribution if

(a) $n \longrightarrow \infty$ and $p \longrightarrow 0$ (b) $n \longrightarrow 0$ and $p \longrightarrow q$

(c) $n \longrightarrow \infty$ and $p \longrightarrow n$ (d) $n \longrightarrow \infty$ neither p nor q is small

13. Skewness and Kurtosis of $N(\mu, \sigma^2)$ are

(a) 0, 1 (b) 0, 3 (c) 0, 2 (d) 0, 0

14. In $N(\mu, \sigma^2)$ the value of $P(|X - \mu| < 2\sigma)$ is

(a) 0.9544 (b) 0.6826 (c) 0.9973 (d) 0.0027

15. If $X \sim N(6, 1.2)$ and $P(0 < Z < 1) = 0.3413$ then $P(4.8 < X < 7.2)$ is

(a) 0.3413 (b) 0.6587 (c) 0.6826 (d) 0.3174

II. Fill in the blanks:

16. Number of trials in a Bernoulli distribution is ----------

17. If the mean and variance of a binomial distribution are 8 and 4 respectively then $P(X = 1)$ is ----------

18. The trials in a Binomial distribution are ----------

19. If $X \sim U(a, b)$ then its variance is ----------

20. ---------- represents a Standard Normal distribution?

III. Answer shortly:

21. State the characteristics of a Bernoulli distribution.

22. State the conditions of a Binomial distribution.

23. Comment on: "the mean and the standard deviation of a Binomial distribution are 7 and 4".

24. For the Binomial distribution $(0.68 + 0.32)^{10}$ find the probability of two successes.

25. If on an average 8 ships out of 10 arrive safely at a port find the mean and standard deviation of the number of ships arriving safely out of 1600 ships.

26. Give any two examples of a Poisson distribution.

27. The variance of a Poisson distribution is 0.5. Find P(X=3) [e-0.5 = 0.6065]

28. State the characteristics of a Poisson distribution.